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MAM1019H

Hand-in 5

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Answers

1. (a) False
(b) True
(c) False
(d) True
2. Suppose $a \in A$ is a minimal element of a . Thus there is no $x \in A, x \neq a$ and $x \leq a$. But $(P, \leq)$ is a linear order, so $\forall x, y \in P$, we either have that $x \leq y$ or $y \leq x$. Thus we must have that $\forall x \in A \subseteq P, x \leq a$ or $a \leq x$. For a to be minimal, we must have that $a \leq x \quad \forall x \in A$. Therefore a is the minimum of A .
3. Consider $f : P \rightarrow \mathbb{N}$ defined by: $f(x) = \log_2 x$. Note that $\forall x \in P, \exists n \in \mathbb{N}, x = 2^n$ by the definition of P . Thus $f(x) = \log_2 2^n = n \in \mathbb{N}$ so this is well defined. Moreover, for any $n \in \mathbb{N}, 2^n \in P$ by the definition of P and $f(2^n) = \log_2 2^n = n$ so f is surjective. Now if $f(x) = f(x')$ for some $x, x' \in P$, then $\log_2 x = \log_2 x' \Rightarrow 2^{\log_2 x} = 2^{\log_2 x'} \Rightarrow x = x'$ so f is injective. Thus f is a bijection. Now we need to show that for any $p, p' \in P, p|p' \iff f(p) \leq f(p')$.
($\Rightarrow$) Assume $p, p' \in P$ such that $p|p'$. Since $p, p' \in \mathbb{N}$, we have that $np = p', n \in \mathbb{N} \Rightarrow p \leq p' \Rightarrow \log_2 p \leq \log_2 p' \Rightarrow f(p) \leq f(p')$. Thus $p|p' \Rightarrow f(p) \leq f(p')$.
($\Leftarrow$) Now assume that $f(p) \leq f(p')$ for some $p, p' \in P$. Thus $\log_2 p \leq \log_2 p'$. Now, by the definition of $P, \exists n, n' \in \mathbb{N}$ such that $p = 2^n$ and $p' = 2^{n'}$. Thus, by our inequality, $\log_2 2^n \leq \log_2 2^{n'} \Rightarrow n \leq n'$. But if $n \leq n'$, then $n' = n + m$ for some $m \in \mathbb{N}$ meaning $2^{n'} = 2^{n+m}$ for some $m \in \mathbb{N}$, so $2^{n'} = 2^n \cdot 2^m \Rightarrow 2^n | 2^{n'} \Rightarrow p|p'$ so $f(p) \leq f(p') \Rightarrow p|p'$. Thus $p|p' \iff f(p) \leq f(p')$ making f an order isomorphism.